



SYSTEM: 7ep

system: "7ep"

services:

- name: "Authentication Service"

responsibilities:

- "Register librarians with username and password"
- "Enforce strong password policies"
- "Authenticate librarians with username and password"
- "Validate librarian sessions"
- "Handle access denials securely"

dependencies:

- "Audit Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Book Catalog Service"

responsibilities:

- "Register books in collection by title"
- "Prevent duplicate book registrations"
- "Delete registered books"
- "List all registered books"
- "Consult books by identifier or title"
- "List books available for loan"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/book-catalog-service"

- name: "Reader Management Service"

responsibilities:

- "Register readers eligible for loans"
- "Prevent duplicate reader registrations"
- "Delete registered readers"
- "List all registered readers"
- "Consult readers by identifier or name"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/reader-management-service"

- name: "Loan Management Service"

responsibilities:

- "Register book loans with loan dates"
- "Ensure books are loaned to only one reader at a time"
- "Allow readers to borrow multiple books"
- "Remove loan links when books or readers are deleted"
- "Preserve consistency among books readers and loans"

dependencies:

- "Book Catalog Service"
- "Reader Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/loan-management-service"

- name: "Environment Management Service"

responsibilities:

- "Clear data and base structure to reset environment"
- "Recreate base structure"
- "Restore environment to clean state"
- "Coordinate functional areas for consistent environment reset"

dependencies:

- "Authentication Service"
- "Book Catalog Service"
- "Reader Management Service"
- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/environment-management-service"

- name: "Demo Utility Service"

responsibilities:

- "Sum two integers"
- "Calculate Fibonacci sequence values using multiple approaches"
- "Calculate Ackermann function results using multiple approaches"
- "Provide service catalog for testing endpoints"

dependencies:

- ""

suggested_endpoints:

- "/api/demo-utility-service"

- name: "Audit Service"

responsibilities:

- "Record main system operations for traceability"
- "Maintain central decoupled audit event logs"

dependencies:

- ""

suggested_endpoints:

- "/api/audit-service"

interactions:

- from: "Authentication Service"
to: "Audit Service"
- from: "Book Catalog Service"
to: "Loan Management Service"
- from: "Book Catalog Service"
to: "Audit Service"
- from: "Reader Management Service"
to: "Loan Management Service"
- from: "Reader Management Service"
to: "Audit Service"
- from: "Loan Management Service"
to: "Book Catalog Service"
- from: "Loan Management Service"
to: "Reader Management Service"
- from: "Loan Management Service"
to: "Audit Service"
- from: "Environment Management Service"
to: "Authentication Service"
- from: "Environment Management Service"
to: "Book Catalog Service"
- from: "Environment Management Service"
to: "Reader Management Service"
- from: "Environment Management Service"
to: "Loan Management Service"
- from: "Environment Management Service"
to: "Audit Service"

SYSTEM: acmeair

system: "acmeair"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate customer credentials"
- "Create and maintain customer usage sessions"
- "Validate active customer sessions"
- "Invalidate sessions immediately upon logout"
- "Expire sessions automatically after configured period"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"
- name: "Flight Search Service"
- responsibilities:
 - "Search one-way and round-trip flights by origin airport destination airport and travel date"
 - "Present flight options with departure arrival and reference fares"
 - "Manage registered airports routes schedules and reference fares"
- dependencies:
 - ""
- suggested_endpoints:
 - "/api/flight-search-service"
- name: "Booking Service"
- responsibilities:
 - "Create one-way and round-trip flight bookings"
 - "Generate unique booking identifiers and record booking dates"
 - "Consult and list completed customer bookings"
 - "Cancel selected bookings and update active booking status"
- dependencies:
 - "Authentication Service"
 - "Flight Search Service"
- suggested_endpoints:
 - "/api/booking-service"
- name: "Customer Profile Service"
- responsibilities:
 - "Retrieve customer profile contact and address information"
 - "Update customer contact and address details"
 - "Require current password to confirm profile updates"
 - "Protect customer personal data against unauthorized access"
- dependencies:
 - "Authentication Service"
- suggested_endpoints:
 - "/api/customer-profile-service"
- name: "Loyalty Service"
- responsibilities:
 - "Consult customer loyalty program status"
 - "Present annual accumulated miles as read-only data"
 - "Present total accumulated miles as read-only data"
- dependencies:
 - "Authentication Service"
- suggested_endpoints:
 - "/api/loyalty-service"
- name: "Environment Administration Service"
- responsibilities:
 - "Execute initial load and reload of application operational database"

- "Configure and generate sample customer accounts and operational data"
- "Provide summarized operational indicators and metrics without direct database access"
- "Supervise and reset operational data across capabilities"

dependencies:

- "Authentication Service"
- "Flight Search Service"
- "Booking Service"
- "Customer Profile Service"

suggested_endpoints:

- "/api/environment-administration-service"

interactions:

- from: "Booking Service"
to: "Authentication Service"
- from: "Booking Service"
to: "Flight Search Service"
- from: "Customer Profile Service"
to: "Authentication Service"
- from: "Loyalty Service"
to: "Authentication Service"
- from: "Environment Administration Service"
to: "Authentication Service"
- from: "Environment Administration Service"
to: "Flight Search Service"
- from: "Environment Administration Service"
to: "Booking Service"
- from: "Environment Administration Service"
to: "Customer Profile Service"

SYSTEM: Cargo-Tracker

system: "Cargo-Tracker"

services:

- name: "Cargo Management Service"

responsibilities:

- "Register new cargo with origin destination and arrival deadline"
- "Assign unique tracking identifier to cargo"
- "Validate cargo origin and destination from registered logistics locations"
- "Maintain cargo lifecycle and origin immutability"
- "Update cargo destination and trigger route reevaluation"
- "Assign selected route to cargo"

dependencies:

- "Route Planning Service"

suggested_endpoints:

- "/api/cargo-management-service"

- name: "Route Planning Service"

responsibilities:

- "Retrieve and present candidate routes compatible with cargo needs"
- "Validate candidate routes against origin destination and arrival deadline"
- "Provide route leg details including load points unload points voyages and expected dates"
- "Reevaluate and replan routes for cargos with changed destinations or off-route exceptions"

dependencies:

- "Cargo Management Service"

suggested_endpoints:

- "/api/route-planning-service"

- name: "Logistics Event Registration Service"

responsibilities:

- "Register logistics events including reception load unload customs and pickup"
- "Validate event associations with cargo location and voyage"
- "Reject invalid registration attempts and isolate rejected records"
- "Record occurrence and registration timestamps for auditability"
- "Recalculate transport status last known location and plan adherence"
- "Detect off-route execution and mark cargo arrival or completion"

dependencies:

- "Cargo Management Service"
- "Route Planning Service"

suggested_endpoints:

- "/api/logistics-event-registration-service"

- name: "Cargo Tracking Service"

responsibilities:

- "Consult cargo current situation and operational status by tracking identifier"
- "Consolidate last known location next expected step and estimated arrival"
- "Maintain and display logistics event history distinguishing expected and unexpected events"
- "Signal off-route operational exceptions and deviations"
- "Provide geographic visualization of origin final destination and last known location"
- "Distinguish delivered in-transit and deviated cargos"

dependencies:

- "Cargo Management Service"
- "Route Planning Service"
- "Logistics Event Registration Service"

suggested_endpoints:

- "/api/cargo-tracking-service"

- name: "Integration Service"

responsibilities:

- "Receive logistics events across operational interfaces integration services and periodic file inputs"
- "Process input files asynchronously and separate invalid records"
- "Provide consolidated cargo views and standardized data formats for external and monitoring channels"
- "Ensure tracking model consistency across all channels"

dependencies:

- "Logistics Event Registration Service"
- "Cargo Tracking Service"

suggested_endpoints:

- "/api/integration-service"

interactions:

- from: "Cargo Management Service"
to: "Route Planning Service"
- from: "Route Planning Service"
to: "Cargo Management Service"
- from: "Logistics Event Registration Service"
to: "Cargo Management Service"
- from: "Logistics Event Registration Service"
to: "Route Planning Service"
- from: "Cargo Tracking Service"
to: "Cargo Management Service"
- from: "Cargo Tracking Service"
to: "Route Planning Service"
- from: "Cargo Tracking Service"
to: "Logistics Event Registration Service"
- from: "Integration Service"
to: "Logistics Event Registration Service"
- from: "Integration Service"
to: "Cargo Tracking Service"

SYSTEM: DayTrader7

system: "DayTrader7"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate investor credentials"
- "Manage investor login and logout sessions"
- "Enforce access control to restricted investor resources"
- "Ensure operations executed during sessions are associated with authenticated investors"

dependencies:

- "Investor Account Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Investor Account Service"

responsibilities:

- "Register new investor accounts"
- "Validate investor registration and profile update fields"
- "Maintain investor financial account and cash balance"

- "Update investor profile information"
- "Process balance debits and credits resulting from trading operations"

dependencies:

- "Authentication Service"
- "Trading Service"
- "Portfolio Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/investor-account-service"

- name: "Market Data Service"

responsibilities:

- "Query asset quotes individually and simultaneously"
- "Provide consolidated market summary and market indicators"
- "Handle invalid asset codes during queries"
- "Update traded volume and market indicators following trading operations"

dependencies:

- "Trading Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/market-data-service"

- name: "Trading Service"

responsibilities:

- "Create and manage buy and sell trading orders"
- "Apply fixed operation fees to trading orders"
- "Track trading order lifecycle and status"
- "Execute orders immediately or deferred based on simulation configuration"
- "Maintain investor order history and recent orders summary"
- "Generate completed order alerts"
- "Cancel sell orders for unavailable positions"

dependencies:

- "Investor Account Service"
- "Portfolio Service"
- "Market Data Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/trading-service"

- name: "Portfolio Service"

responsibilities:

- "Maintain investor portfolio positions"
- "Register new positions from completed buy orders"
- "Remove positions upon completed sell orders"
- "Consult portfolio positions and calculate performance metrics"
- "Generate consolidated financial position summary"

- "Initiate asset sales directly from portfolio positions"

dependencies:

- "Trading Service"
- "Market Data Service"
- "Investor Account Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/portfolio-service"

- name: "Simulation Administration Service"

responsibilities:

- "Configure simulation operational parameters"
- "Manage order processing modes and alert preferences"
- "Prepare and rebuild simulation database"
- "Reset simulation operational data"
- "Provide summarized execution statistics and operational monitoring indicators"

dependencies:

- "Investor Account Service"
- "Market Data Service"
- "Trading Service"
- "Portfolio Service"

suggested_endpoints:

- "/api/simulation-administration-service"

interactions:

- from: "Authentication Service"
to: "Investor Account Service"
- from: "Investor Account Service"
to: "Authentication Service"
- from: "Investor Account Service"
to: "Trading Service"
- from: "Investor Account Service"
to: "Portfolio Service"
- from: "Investor Account Service"
to: "Simulation Administration Service"
- from: "Market Data Service"
to: "Trading Service"
- from: "Market Data Service"
to: "Simulation Administration Service"
- from: "Trading Service"
to: "Investor Account Service"
- from: "Trading Service"
to: "Portfolio Service"
- from: "Trading Service"
to: "Market Data Service"

- from: "Trading Service"
to: "Simulation Administration Service"
- from: "Portfolio Service"
to: "Trading Service"
- from: "Portfolio Service"
to: "Market Data Service"
- from: "Portfolio Service"
to: "Investor Account Service"
- from: "Portfolio Service"
to: "Simulation Administration Service"
- from: "Simulation Administration Service"
to: "Investor Account Service"
- from: "Simulation Administration Service"
to: "Market Data Service"
- from: "Simulation Administration Service"
to: "Trading Service"
- from: "Simulation Administration Service"
to: "Portfolio Service"

SYSTEM: Jokul

system: "Jokul"

services:

- name: "Authentication Service"

responsibilities:

- "Register users with credentials"
- "Authenticate user login"
- "Manage user sessions"
- "Control user profiles and permissions"
- "Prevent duplicate user registrations"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Catalog Service"

responsibilities:

- "Register new movies"
- "Provide paginated movie listings"
- "Provide movie details and cast"
- "Manage movie categories"
- "List movies by category"
- "Delete movies individually and in batch"
- "Ensure movie title uniqueness"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Media Service"

responsibilities:

- "Upload video files"
- "Store and manage video files in local repository and delivery area"
- "Provide movie playback"
- "Provide video file download links"
- "Verify video availability for playback"

dependencies:

- "Authentication Service"
- "Catalog Service"

suggested_endpoints:

- "/api/media-service"

interactions:

- from: "Catalog Service"
to: "Authentication Service"
- from: "Media Service"
to: "Authentication Service"
- from: "Media Service"
to: "Catalog Service"

SYSTEM: JpetStore

system: "JpetStore"

services:

- name: "Catalog Service"

responsibilities:

- "Browse main catalog and organize products by animal categories"
- "Provide category navigation paths"
- "View product lists and item details"
- "Search products by keyword and validate search inputs"

dependencies:

- "Inventory Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Customer Account Service"

responsibilities:

- "Register customer identification contact and address data"
- "Manage customer profile and account preferences"
- "Configure preferred language and favorite animal category"
- "Enable personalized product lists and promotional banners based on favorite category"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/customer-account-service"

- name: "Authentication Service"

responsibilities:

- "Authenticate customer credentials using username and password"
- "Manage customer usage sessions and session termination"
- "Control and protect authenticated access to features"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Shopping Cart Service"

responsibilities:

- "Add and consolidate items in shopping cart"
- "View cart with items quantities unit prices and subtotal"
- "Update item quantities and remove items from cart"
- "Clear customer cart upon purchase completion"

dependencies:

- "Catalog Service"
- "Inventory Service"

suggested_endpoints:

- "/api/shopping-cart-service"

- name: "Inventory Service"

responsibilities:

- "Track item stock availability and backorder status"
- "Provide updated stock information at critical purchase journey points"
- "Register stock decrease upon order completion"

dependencies:

- "Catalog Service"

suggested_endpoints:

- "/api/inventory-service"

- name: "Order Service"

responsibilities:

- "Initiate and process checkout using customer registration data"
- "Collect payment information and card type"
- "Manage billing and delivery addresses"
- "Finalize orders and generate unique purchase identifiers"
- "Record purchase date items quantities values and status"
- "Consult completed order lists and order details with access restrictions"

dependencies:

- "Authentication Service"
- "Customer Account Service"

- "Shopping Cart Service"

- "Inventory Service"

suggested_endpoints:

- "/api/order-service"

interactions:

- from: "Catalog Service"

- to: "Inventory Service"

- from: "Customer Account Service"

- to: "Authentication Service"

- from: "Shopping Cart Service"

- to: "Catalog Service"

- from: "Shopping Cart Service"

- to: "Inventory Service"

- from: "Inventory Service"

- to: "Catalog Service"

- from: "Order Service"

- to: "Authentication Service"

- from: "Order Service"

- to: "Customer Account Service"

- from: "Order Service"

- to: "Shopping Cart Service"

- from: "Order Service"

- to: "Inventory Service"

SYSTEM: Pet-Clinic

system: "Pet-Clinic"

services:

- name: "Owner Service"

responsibilities:

- "Register owners"

- "Update owner data"

- "Search owners by last name"

- "Retrieve owner details and paginated owner lists"

dependencies:

- ""

suggested_endpoints:

- "/api/owner-service"

- name: "Animal Service"

responsibilities:

- "Register animals linked to owners"

- "Update animal data"

- "Validate animal birth date and name uniqueness per owner"

- "Provide predefined animal types"

dependencies:

- "Owner Service"

suggested_endpoints:

- "/api/animal-service"

- name: "Visit Service"

responsibilities:

- "Register animal visits"
- "Maintain visit and service history"
- "Associate visits with animals and owners"
- "Resume pending internal visit processes"

dependencies:

- "Animal Service"
- "Veterinarian Service"

suggested_endpoints:

- "/api/visit-service"

- name: "Veterinarian Service"

responsibilities:

- "List veterinarians and specialties"
- "Assign visits to veterinarians based on workload"
- "Calculate veterinarian workload and next expected visits"
- "Provide structured veterinarian data"

dependencies:

- "Visit Service"

suggested_endpoints:

- "/api/veterinarian-service"

interactions:

- from: "Animal Service"
to: "Owner Service"
- from: "Visit Service"
to: "Animal Service"
- from: "Visit Service"
to: "Veterinarian Service"
- from: "Veterinarian Service"
to: "Visit Service"

SYSTEM: TNTConcept

system: "TNTConcept"

services:

- name: "Administration and Configuration Service"

responsibilities:

- "Maintain support records including departments"

dependencies:

- "contract types"

suggested_endpoints:

- "/api/administration-and-configuration-service"

- name: "Identity and Access Service"

responsibilities:

- "Register and maintain users with personal"

dependencies:

- "contractual"

suggested_endpoints:

- "/api/identity-and-access-service"

- name: "Workforce Management Service"

responsibilities:

- "Maintain work agreements"
- "Manage vacation requests and request states"
- "Calculate vacation balances"
- "Monitor contract expiration"

dependencies:

- "Administration and Configuration Service"
- "Identity and Access Service"
- "Project and Work Tracking Service"
- "Commissioning Service"
- "Notification Service"

suggested_endpoints:

- "/api/workforce-management-service"

- name: "Project and Work Tracking Service"

responsibilities:

- "Maintain projects linked to clients"
- "Register internal and external activities"
- "Classify activities and calculate worked hours"
- "Support evidence policies"

dependencies:

- "Identity and Access Service"
- "Workforce Management Service"
- "CRM and Proposals Service"
- "Financial Management Service"
- "Billing Service"
- "Commissioning Service"
- "Notification Service"
- "File Storage Service"

suggested_endpoints:

- "/api/project-and-work-tracking-service"

- name: "CRM and Proposals Service"

responsibilities:

- "Maintain organizations"

dependencies:

- "contacts"

suggested_endpoints:

- "/api/crm-and-proposals-service"

- name: "Financial Management Service"

responsibilities:

- "Maintain accounts and financial entries"
- "Calculate and track financial indicators"
- "Import billable hours and costs"

dependencies:

- "Billing Service"
- "Project and Work Tracking Service"

suggested_endpoints:

- "/api/financial-management-service"

- name: "Billing Service"

responsibilities:

- "Detail invoices"
- "Calculate totals"

dependencies:

- "taxes"

suggested_endpoints:

- "/api/billing-service"

- name: "Content and Document Management Service"

responsibilities:

- "Organize documents by categories and versions"
- "Associate documents with users and areas"

dependencies:

- "Identity and Access Service"
- "File Storage Service"

suggested_endpoints:

- "/api/content-and-document-management-service"

- name: "Commissioning Service"

responsibilities:

- "Register and manage orders and commissions"

dependencies:

- "Workforce Management Service"
- "Project and Work Tracking Service"
- "Notification Service"
- "File Storage Service"

suggested_endpoints:

- "/api/commissioning-service"

- name: "Reporting Service"

responsibilities:

- "Consolidate data across business domains"
- "Generate and export reports"

dependencies:

- "Workforce Management Service"
- "Project and Work Tracking Service"
- "CRM and Proposals Service"
- "Financial Management Service"
- "Billing Service"
- "Commissioning Service"

suggested_endpoints:

- "/api/reporting-service"

- name: "Notification Service"

responsibilities:

- "Send email notifications"
- "Process notification requests from domain services"

dependencies:

- "Administration and Configuration Service"
- "Workforce Management Service"
- "Project and Work Tracking Service"
- "Commissioning Service"

suggested_endpoints:

- "/api/notification-service"

- name: "File Storage Service"

responsibilities:

- "Store and retrieve files"
- "Manage storage configurations"

dependencies:

- "Administration and Configuration Service"
- "Project and Work Tracking Service"
- "Billing Service"
- "Content and Document Management Service"
- "Commissioning Service"

suggested_endpoints:

- "/api/file-storage-service"

interactions:

- from: "Administration and Configuration Service"
to: "contract types"
- from: "Identity and Access Service"
to: "contractual"
- from: "Workforce Management Service"
to: "Administration and Configuration Service"
- from: "Workforce Management Service"
to: "Identity and Access Service"

- from: "Workforce Management Service"
to: "Project and Work Tracking Service"

- from: "Workforce Management Service"
to: "Commissioning Service"

- from: "Workforce Management Service"
to: "Notification Service"

- from: "Project and Work Tracking Service"
to: "Identity and Access Service"

- from: "Project and Work Tracking Service"
to: "Workforce Management Service"

- from: "Project and Work Tracking Service"
to: "CRM and Proposals Service"

- from: "Project and Work Tracking Service"
to: "Financial Management Service"

- from: "Project and Work Tracking Service"
to: "Billing Service"

- from: "Project and Work Tracking Service"
to: "Commissioning Service"

- from: "Project and Work Tracking Service"
to: "Notification Service"

- from: "Project and Work Tracking Service"
to: "File Storage Service"

- from: "CRM and Proposals Service"
to: "contacts"

- from: "Financial Management Service"
to: "Billing Service"

- from: "Financial Management Service"
to: "Project and Work Tracking Service"

- from: "Billing Service"
to: "taxes"

- from: "Content and Document Management Service"
to: "Identity and Access Service"

- from: "Content and Document Management Service"
to: "File Storage Service"

- from: "Commissioning Service"
to: "Workforce Management Service"

- from: "Commissioning Service"
to: "Project and Work Tracking Service"

- from: "Commissioning Service"
to: "Notification Service"

- from: "Commissioning Service"
to: "File Storage Service"

- from: "Reporting Service"

to: "Workforce Management Service"

- from: "Reporting Service"

to: "Project and Work Tracking Service"

- from: "Reporting Service"

to: "CRM and Proposals Service"

- from: "Reporting Service"

to: "Financial Management Service"

- from: "Reporting Service"

to: "Billing Service"

- from: "Reporting Service"

to: "Commissioning Service"

- from: "Notification Service"

to: "Administration and Configuration Service"

- from: "Notification Service"

to: "Workforce Management Service"

- from: "Notification Service"

to: "Project and Work Tracking Service"

- from: "Notification Service"

to: "Commissioning Service"

- from: "File Storage Service"

to: "Administration and Configuration Service"

- from: "File Storage Service"

to: "Project and Work Tracking Service"

- from: "File Storage Service"

to: "Billing Service"

- from: "File Storage Service"

to: "Content and Document Management Service"

- from: "File Storage Service"

to: "Commissioning Service"